THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 13	13 has only two factors, 1 and 13.
2	**(a)** 27	27 has more than two factors.
3	**(b)** 6	Factors: 1, 2, 3, 4, 6, 12.
4	**(c)** 42	7×6
5	**(c)** 363	Divisibility test of 3.
6	**(d)** 10	Add the primes 2, 3, 5, 7, ... (1 is not prime).
7	**(b)** 2	Definition
8	**(a)** 29	29 has only two factors, 1 and 29.
9	**(b)** 49	49 has more than two factors.
10	**(d)** 6	Factors: 1, 2, 3, 6, 9, 18.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Find pairs whose product is 16 (1); Factors: 1, 2, 4, 8, 16 (1).	1+1
12	Multiply 6 by 1, 2, 3, 4, 5 (1); 6, 12, 18, 24, 30 (1).	1+1
13	Factors of 31: 1, 31 (1); Prime: only two factors (1).	1+1
14	Find pairs whose product is 24 (1); Factors: 1, 2, 3, 4, 6, 8, 12, 24 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Divide by primes in turn (1); $36 = 2 \times 2 \times 3 \times 3$ (1); Only prime factors appear (1).	1+1+1
16	By 2: no (last digit 5) (1); By 3: yes (digit sum 9) (1); By 5: yes (1).	1+1+1
17	Test each number for factors (1); 11, 13, 17, 19, 23, 29 (2).	1+2
	OR: Divide by primes in turn (1); $48 = 2 \times 2 \times 2 \times 2 \times 3$ (1); Only prime factors appear (1).	1+1+1
18	By 2: yes (last digit 6) (1); By 3: yes (digit sum 12) (1); By 5: no (1).	1+1+1
	OR: Test each number for factors (1); 31, 37, 41, 43, 47 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$12 = 2 \times 2 \times 3$ (1); $18 = 2 \times 3 \times 3$ (2); Common prime factors: 2, 3 (2).	1+2+2
	OR: Factors: 1, 2, 4, 7, 14, 28 (1); Count $= 6$ (1); Sum of all factors $= 56$ (1); Even (2).	1+1+1+2
20	$20 = 2 \times 2 \times 5$ (1); $30 = 2 \times 3 \times 5$ (2); Common prime factors: 2, 5 (2).	1+2+2
	OR: Factors: 1, 2, 3, 5, 6, 10, 15, 30 (1); Count $= 8$ (1); Sum of all factors $= 72$ (1); Even (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 72	9×8
2	**(b)** 435	Divisibility test of 5.
3	**(a)** 17	Add the primes 2, 3, 5, 7, ... (1 is not prime).
4	**(c)** 2	Definition
5	**(b)** 37	37 has only two factors, 1 and 37.
6	**(a)** 51	51 has more than two factors.
7	**(c)** 6	Factors: 1, 2, 4, 5, 10, 20.
8	**(c)** 60	12×5
9	**(a)** 567	Divisibility test of 9.
10	**(c)** 28	Add the primes 2, 3, 5, 7, ... (1 is not prime).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Multiply 8 by 1, 2, 3, 4, 5 (1); 8, 16, 24, 32, 40 (1).	1+1
12	Factors of 39: 1, 3, 13, 39 (1); Composite: more than two factors (1).	1+1
13	Find pairs whose product is 35 (1); Factors: 1, 5, 7, 35 (1).	1+1
14	Multiply 13 by 1, 2, 3, 4, 5 (1); 13, 26, 39, 52, 65 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Divide by primes in turn (1); $60 = 2 \times 2 \times 3 \times 5$ (1); Only prime factors appear (1).	1+1+1
16	By 2: yes (last digit 0) (1); By 3: yes (digit sum 6) (1); By 5: yes (1).	1+1+1
17	Test each number for factors (1); 53, 59, 61, 67 (2).	1+2
	OR: Divide by primes in turn (1); $72 = 2 \times 2 \times 2 \times 3 \times 3$ (1); Only prime factors appear (1).	1+1+1
18	By 2: no (last digit 1) (1); By 3: yes (digit sum 6) (1); By 5: no (1).	1+1+1
	OR: Test each number for factors (1); 71, 73, 79, 83, 89 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$28 = 2 \times 2 \times 7$ (1); $42 = 2 \times 3 \times 7$ (2); Common prime factors: 2, 7 (2).	1+2+2
	OR: Factors: 1, 2, 3, 4, 6, 9, 12, 18, 36 (1); Count $= 9$ (1); Sum of all factors $= 91$ (1); Even (2).	1+1+1+2
20	$15 = 3 \times 5$ (1); $35 = 5 \times 7$ (2); Common prime factors: 5 (2).	1+2+2
	OR: Factors: 1, 3, 5, 9, 15, 45 (1); Count $= 6$ (1); Sum of all factors $= 78$ (1); Odd (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** neither prime nor composite	Definition
2	**(a)** 53	53 has only two factors, 1 and 53.
3	**(c)** 63	63 has more than two factors.
4	**(d)** 6	Factors: 1, 2, 4, 7, 14, 28.
5	**(a)** 105	15×7
6	**(d)** 90	Divisibility test of 2.
7	**(a)** 41	Add the primes 2, 3, 5, 7, ... (1 is not prime).
8	**(d)** 71	71 has only two factors, 1 and 71.
9	**(a)** 91	91 has more than two factors.
10	**(b)** 8	Factors: 1, 2, 3, 5, 6, 10, 15, 30.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Factors of 57: 1, 3, 19, 57 (1); Composite: more than two factors (1).	1+1
12	Find pairs whose product is 42 (1); Factors: 1, 2, 3, 6, 7, 14, 21, 42 (1).	1+1
13	Multiply 17 by 1, 2, 3, 4, 5 (1); 17, 34, 51, 68, 85 (1).	1+1
14	Factors of 61: 1, 61 (1); Prime: only two factors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Divide by primes in turn (1); $84 = 2 \times 2 \times 3 \times 7$ (1); Only prime factors appear (1).	1+1+1
16	By 2: yes (last digit 0) (1); By 3: no (digit sum 7) (1); By 5: yes (1).	1+1+1
17	Test each number for factors (1); 23, 29, 31, 37 (2).	1+2
	OR: Divide by primes in turn (1); $90 = 2 \times 3 \times 3 \times 5$ (1); Only prime factors appear (1).	1+1+1
18	By 2: no (last digit 5) (1); By 3: yes (digit sum 18) (1); By 5: yes (1).	1+1+1
	OR: Test each number for factors (1); 41, 43, 47, 53, 59 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$24 = 2 \times 2 \times 2 \times 3$ (1); $40 = 2 \times 2 \times 2 \times 5$ (2); Common prime factors: 2 (2).	1+2+2
	OR: Factors: 1, 2, 5, 10, 25, 50 (1); Count $= 6$ (1); Sum of all factors $= 93$ (1); Even (2).	1+1+1+2
20	$45 = 3 \times 3 \times 5$ (1); $60 = 2 \times 2 \times 3 \times 5$ (2); Common prime factors: 3, 5 (2).	1+2+2
	OR: Factors: 1, 2, 4, 7, 8, 14, 28, 56 (1); Count $= 8$ (1); Sum of all factors $= 120$ (1); Even (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.